

Situation	Task	Actions	Results	Lesson Learned
The context is about Yield 3.0 project, which is a yield monitor system to be installed in transition housing of a grain tank. Please see the illustration, this was a long-awaited project with goals of accurate yield sensing, John Deere IP ownership, cost reduction, and eliminate supplier dependency (ag leader). It required estimated capital investment of 360K. It came to me directly at DB phase in 2021 from another quality engineer. This project had gone through many adulations before coming to me. There was a prototype made in FB which had multiple failure modes.	The quality manager at that time trained me on various phase deliverables and duties as a core QE or now it is called as Program QE. My quality engineers on electrical system and in grain handling area were good. I realized that the senior electrical engineer who owned the design aspect did not have mechanical design experience in John Deere.. Also there was no mechanical design engineer assigned to this project from grain handling. A staff engineer was working as a consultant so all things were to be decided by Electrical Design Engineer and that's where my coaching opportunity lied. My task was quality integration and complete the project successfully through all phases as a core QE, aligning and guiding the cross functional team.	I brought system clarity and alignment by introducing a system level approach, created boundary diagrams, identified missing requirements and aligned mechanical, sensing and software interfaces . I remember clearly giving analogy of palm, forearm, arm, should, and scapula to the senior engineer to think this project as a system. I led cross functional team to resolve multiple sequentially failures, such as wear plate and bolts by identifying the need for material change, Bracket failures by getting FEA and structural reinforcement in place and sensor accuracy issues resolved by collaborating with ISG in addition to onboarding a data scientist for algorithm refinement. I managed late-stage crises before SOP due to sensor plate failure because of unknown higher than tested mass flow rates for X9 machines. I led Quality Planning, managed program risks and mitigation plans, provide progress updates to stakeholders like core teams, PLMT, Product Council, and Factory Leadership throughout the 4 phases of the EPDP.	I was able to successfully launch the program in MY25, that achieved ~4% accuracy in medium flow crops comparable to previous system, delivered ~\$260 per machine cost savings, reduced supplier dependency and built internal IP, managed post launch issues with limited warranty impact (~\$16K, so far 6 claims), strengthened team capability through coaching and system thinking”	My learning from this project was to spend more time on defining roles and responsibilities earlier. escalate risks earlier instead of trying to resolve everything independently specially under SOP pressure. Also even I was Labor grade 7, I took the projet upon me to take to end and worked with higher ups, gave me confidence